Answer key

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Module 01: Introduction to Data Centre Electrical Design

1.1: 3,608 A. Working: I = 2,500,000 / (sqrt3 x 400)

1.2: 1,897 kVA. Working: S = sqrt(1800^2 + 600^2)

Module 02: IT Load, Facility Load and PUE

2.1: 127.7 kW. Working: P_loss = 2000 x (100/94 - 1)

2.2: 1.3711 -. Working: Re-run the load model with efficiency 94 %: the UPS loss rises and the distribution losses rise with it.

Module 03: Electrical Architecture, A/B Paths and Redundancy

3.1: 71.6 %. Working: loading = load / (1 x 4000 x 1.0)

3.2: 3,200 kVA. Working: (3 - 1) x 2000 x 0.8

Module 04: MV Distribution and Transformer Sizing

4.1: 3,352 kVA. Working: S = 2866 / (0.90 x 0.95)

4.2: 65.6 kA. Working: Z_T = K_T x 0.06 x 400^2 / 2.5e6; I = 1.05 x 400 / (sqrt3 x Z_T)

Module 05: Generator Sizing and Transient Performance

5.1: 1,600 kW. Working: 2000 x 0.8

5.2: 4 units. Working: Needed running units = ceil(2687 / (1600 x 0.8)) = 3; add one spare for N+1.

Module 06: UPS Architecture and Battery Sizing

6.1: 360.0 kWh. Working: E = 1800 x 12 / 60

6.2: 542.8 kWh. Working: E_nom = 360 / 0.96 / (0.8 x 0.95) x 1.1

Module 07: LV Distribution and Switchgear Selection

7.1: 4,596 A. Working: I = 4136 / 0.9

7.2: FAIL. Working: 65 < 74

Module 08: Cable Sizing, Ampacity and Voltage Drop

8.1: 0.1832 mohm/m. Working: R = 0.017241 x (1 + 0.00393 x 70) / 120

8.2: 0.5086 %. Working: dV = sqrt3 x 300 x 40 x (0.00007 x 0.9 + 0.00008 x 0.4359); divide by 400

Module 09: Short-Circuit Fundamentals and Protection Coordination

9.1: 98.8 kA. Working: kappa = 1.02 + 0.98 exp(-0.3); ip = kappa sqrt2 x 40

9.2: 10.5 kA. Working: Ik'' = S / (sqrt3 U) because Z = c U^2 / S

Module 10: Earthing, Bonding, Lightning and Power Quality

10.1: 0.9586 ohm. Working: Rg = 100 [1/1200 + 1/sqrt(50000) (1 + 1/(1 + 0.5 sqrt(20/2500)))]

10.2: 1 %. Working: mean = 400; max deviation = 4; 4/400 = 1 %

Module 11: Single-Line Diagrams, Schedules and Engineering Reports

11.1: NOT VERIFIED. Working: A pass on unverified inputs is not evidence the design works.

11.2: WARNING. Working: A preliminary exceedance is still reported.

Module 12: Commissioning, FAT, SAT and Integrated Systems Testing

12.1: 800.0 kW. Working: 0.75 x 1600 - 400

12.2: 225.0 kWh. Working: E = 1500 x 9 / 60

Question bank answers

Id	Answer	Explanation
Q01.1	B. The UPS bridges the time until generators accept load	The UPS and battery carry the load during the gap.
Q01.2	B. The most onerous failure the design is required to survive	It is the most onerous failure the project has decided is credible.
Q01.3	B. Transformer short-circuit impedance	Impedance comes from the test report.
Q01.4	B. 500 kVA	sqrt(300^2 + 400^2) = 500.
Q02.1	B. Reactive and real power are different quantities; apparent power is the vector sum	S = sqrt(P^2 + Q^2).
Q02.2	B. Efficiency falls	Loss = P x (100/eta - 1).
Q02.3	B. A power ratio at the design point	It is a ratio of power, not annual energy.
Q02.4	B. Not possible: facility power cannot be below IT power	Facility power includes the IT load plus overhead.
Q03.1	B. Two complete independent paths, each able to carry the whole load	Each path can carry the whole critical load.
Q03.2	A. Load carried with the largest unit out	The load must be carried with one unit out.
Q03.3	B. Not independent at that point	A shared point is a common failure.
Q03.4	B. A double contingency, informational unless the project decides it is credible	It is beyond N+1.
Q04.1	B. The derated continuous rating	Derating first.
Q04.2	B. WARNING needing a decision	It needs a recorded decision.
Q04.3	B. A higher fault level	Less impedance passes more fault current.
Q04.4	B. Nothing in continuous sizing	It is not used for continuous loading.
Q05.1	B. Different ISO 8528 rating bases	They are not interchangeable.
Q05.2	B. 1,600	kVA x pf.
Q05.3	A. Steps cause frequency and voltage dips that depend on the unit	Needs manufacturer data.
Q05.4	B. Not automatically good: minimum-load guidance applies	Light running has its own limits.
Q06.1	B. The manufacturer's constant-power discharge data at end of life	Capacity depends on discharge rate.
Q06.2	A. 333 kWh	2000 x 10 / 60.
Q06.3	B. kVA first	kVA = kW / 0.95 is larger.
Q06.4	B. The whole critical load	That is the point of 2N.
Q07.1	B. Design current divided by derating	I / k.
Q07.2	B. The prospective fault level at the busbars	Including the peak current.
Q07.3	B. The assembly manufacturer's verification	The designer checks the evidence.
Q07.4	B. NOT VERIFIED	They are never assumed.
Q08.1	B. Higher	Resistance rises with temperature.
Q08.2	B. The standard for the installation method or the manufacturer	It depends on construction and layout.
Q08.3	B. Divides between runs	Per-run share.
Q08.4	B. Minimum conductor area for a fault of given duration	S >= I sqrt(t) / k.
Q09.1	B. The utility's written statement	It is a site/utility input.

Id	Answer	Explanation
Q09.2	A. X/R	kappa = 1.02 + 0.98 exp(-3 R/X).
Q09.3	B. Device time-current curves and a study	Curves and a validated study.
Q09.4	B. Only the transformer rating and impedance	It assumes an infinite source.
Q10.1	A. Earth-fault current times grid resistance	GPR = Ig x Rg.
Q10.2	C. A detailed mesh and step calculation may still be required	It is only a screening.
Q10.3	B. A complete verified risk assessment and design	Not otherwise.
Q10.4	B. 1 %	4/400.
Q11.1	B. Join drawings, schedules and calculations	The shared key.
Q11.2	B. NOT VERIFIED	A pass is not evidence.
Q11.3	B. The people who approve	Never by the software.
Q11.4	B. A sketch, not evidence	It must trace to something.
Q12.1	B. Factory acceptance test	A unit test at the factory.
Q12.2	B. The whole chain under failure	The chain under failure.
Q12.3	B. Capacity at a known resistive load, not the full behaviour	It is a different load.
Q12.4	B. The specification and manufacturer data	Not from training material.